MAT 221 - CALCULUS I
Lesson 15-16: - Applications of Definite Integrals

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Volume using Cross-Sections

The volume of a solid of integrable cross-sectional area $A(x)$ from $x = a$ and $x = b$ is the integral of $A(x)$ from a to b.

$$V = \int_a^b A(x)\,dx$$

Calculating the Volume of a solid

(1) Sketch the solid and a typical cross-section.

(2) Find a formula for $A(x)$, the area of a typical cross-section.

(3) Find the limits of integration.

(4) Integrate $A(x)$ to find the volume

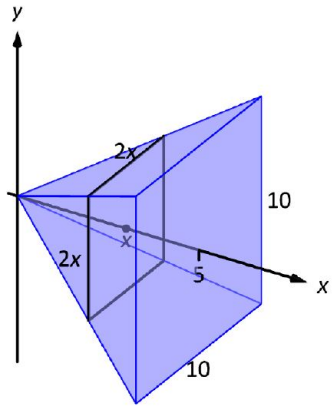
Example 1

(a) A pyramid 5 m high has a square base that is 10 m on one side. The cross section of the pyramid perpendicular to the altitude x m down from the vertex is a square $2x$ m on one side. Find the volume.

(b) A cone 4 m high has a radius 5 m. The cross section of the cone perpendicular to the altitude x m down from the vertex is a circle of radius x m. Find the volume.

Solution

(a) (1) Sketch



(2) The cross section at x is a square $2x$ m on one side, so its area is

$$A(x) = 4x^2.$$

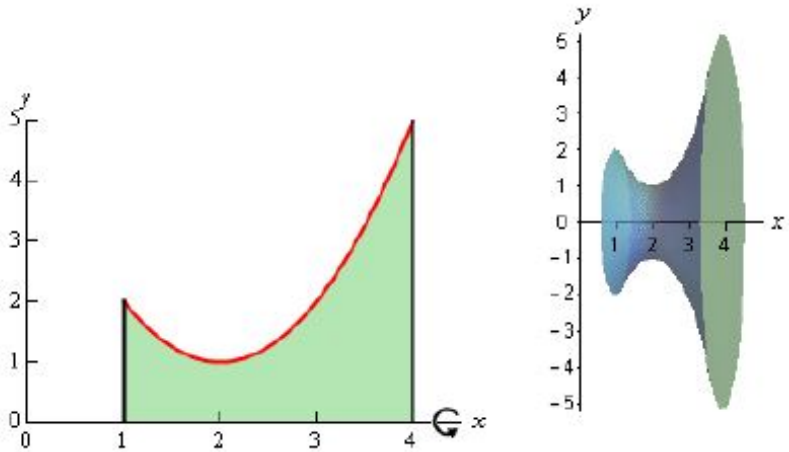
(3) The squares lie on the planes from $x = 0$ to $x = 5$.

(4) $V = \int_0^5 A(x)dx = \int_0^5 4x^2\,dx = \frac{4}{3}x^3|_0^5 = \frac{500}{3}\,m^3$

(b) exercise

Solids of Revolution: The Disk Method

The solid generated by rotating (or revolving) a plane region about an axis in its plane is called a **solid of revolution**. If the curve $y = f(x)$, the $x-$axis and the ordinates at $x = a$ and $x = b$, rotates in a complete revolution about the $x-$axis, it will generate a solid of known volume.



To find the volume of a solid like the one above, we need only observe that the cross-sectional area $A(x)$ is the area of a disk of radius $f(x)$. The area is then

$$A(x) = \pi r^2 = \pi[R(x)]$$

So the definition of volume in this case gives the following theorem.

Theorem 1

If the curve $y = f(x)$ is rotated about the $x-$axis, we have

$$V = \int_a^b A(x)dx = \pi \int_a^b [R(x)]^2 dx = \pi \int_a^b y^2 dx \tag{1}$$

If the curve $y = f(x)$ is rotated about the $y-$axis, we have

$$V = \int_c^d A(y)dy = \pi \int_c^d [R(y)]^2 dy = \pi \int_c^d x^2 dy \tag{2}$$

Example 1

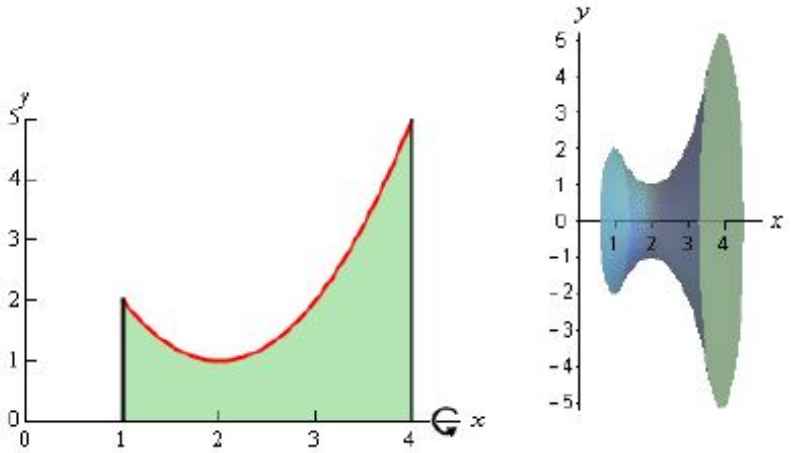
(a) Determine the volume of the solid obtained by rotating the region bounded by $y = x^2 - 4x + 5$, $x = 1$, $x = 4$, and the $x-$axis about the $x-$axis.

(b) Find the volume generated when the curve $y = 5\cos 2x$, the $x-$axis and ordinates at $x = 0$ and $x = \frac{\pi}{4}$ rotates about the $x-$axis through a complete revolution.

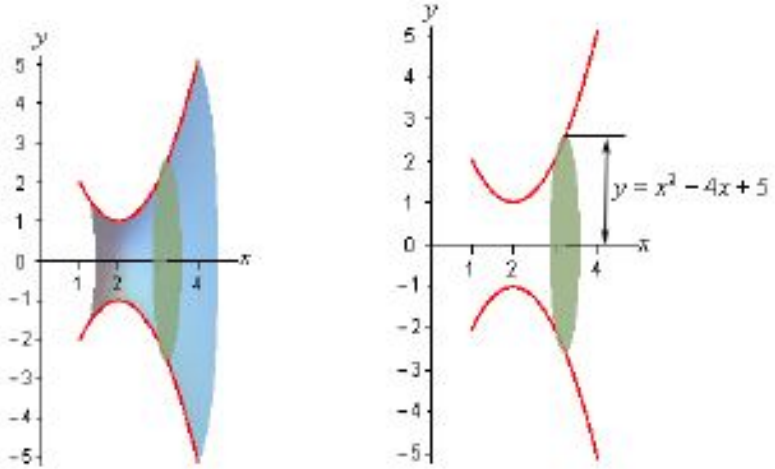
(c) Find the volume of the solid generated by revolving the region between the $y-$axis and the curve $x = \frac{2}{y}$, $1 \leqslant y \leqslant 4$, about the $y-$axis.

Solution

(a) The first thing to do is get a sketch of the bounding region and the solid obtained by rotating the region about the $x-$axis. Here are both the sketches



Below are a couple of sketches showing a typical cross section



We see from the above figure that the radius is simply the distance from the $x-$axis to the curve. The cross-sectional area is then,

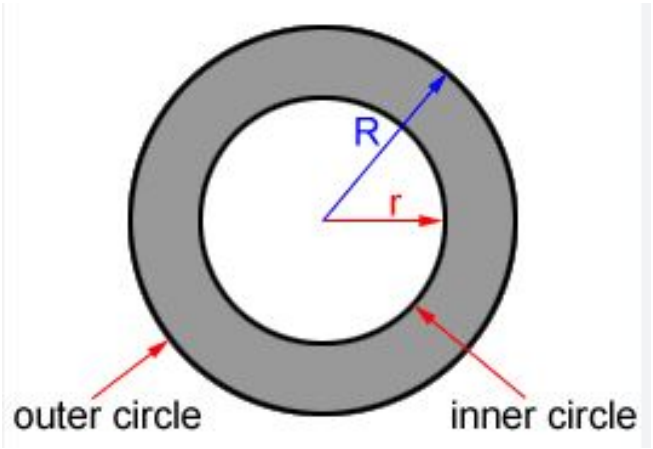
$$A(x) = \pi(x^2 - 4x + 5)^2 = \pi(x^4 - 8x^3 + 26x^2 - 40x + 25)$$

The volume of this solid is then,

$$\begin{aligned} V &= \int_a^b A(x)\,dx \\ &= \pi \int_1^4 (x^4 - 8x^3 + 26x^2 - 40x + 25)\,dx \\ &= \pi \left(\frac{1}{5}x^5 - 2x^4 + \frac{26}{3}x^3 - 20x^2 + 25x \right)_1^4 \\ &= \frac{78\pi}{5} \end{aligned}$$

The Washer (Ring) Method

If the region we revolve to generate a solid does not border on or across the axis of revolution, the solid has a hole in it. The cross-section perpendicular to the axis of revolution are washers/rings.



The washer's area is

$$A(x) = \pi[(R(x))^2 - (r(x))^2].$$

Theorem 2

Volume by washer method for rotation about the $x-$axis

$$V = \int_a^b A(x)dx = \pi \int_a^b [(R(x))^2 - (r(x))^2]dx \tag{3}$$

Theorem 3

Volume by washer method for rotation about the $y-$axis

$$V = \int_c^d A(y)dy = \pi \int_c^d [(R(y))^2 - (r(y))^2]dy \tag{4}$$

Example 2

(a) Determine the volume of the solid obtained by rotating the portion of the region bounded by $y = \sqrt[3]{x}$ and $y = \frac{x}{4}$ that lies in the first quadrant about the $y-$axis.

(b) Determine the volume of the solid obtained by rotating the region bounded by $y = x^2 - 2x$ and $y = x$ about the line $y = 4$.

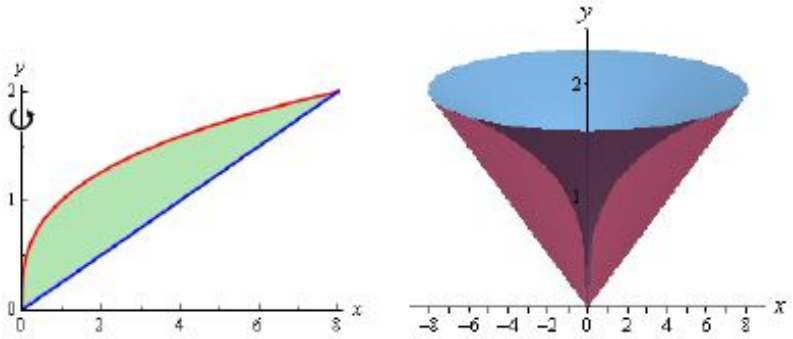
(c) Determine the volume of the solid obtained by rotating the region bounded by $y = 2\sqrt{x-1}$ and $y = x - 1$ about the line $x = -1$.

(d) The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ revolves about the $x-$ axis to generate a solid. Find the volume of the solid.

(e) The region bounded by the parabola $y = x^2$ and the line $y = 2x$ in the first quadrant is revolved about the $y-$axis to generate a solid. Find the volume of the solid.

Solution

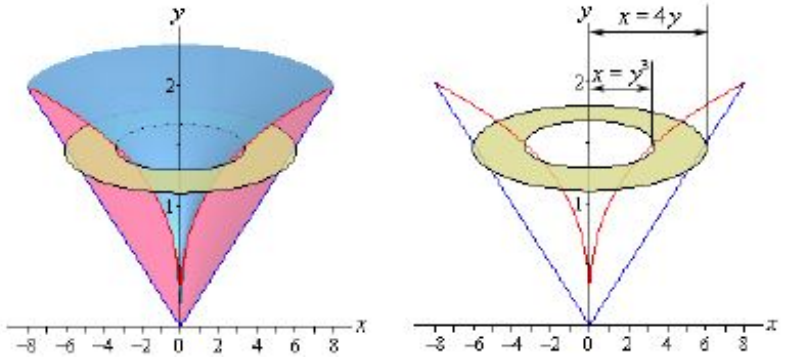
(a) Here are both the sketches.



Because we are revolving about $y-$axis, we then have to rewrite the given functions in terms of x. Here are the functions written in the correct form.

$y = \sqrt[3]{x} \quad \Rightarrow \quad x = y^3$ and $y = \frac{x}{4} \quad \Rightarrow \quad x = 4y.$

Below are the cross-sectional sketches.



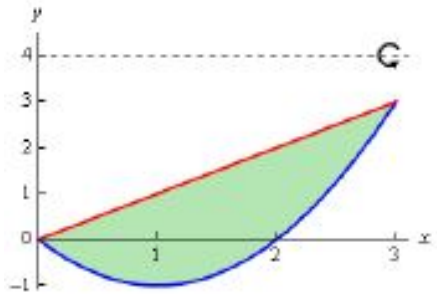
The Cross-sectional area is then,

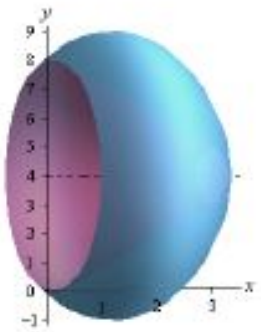
$$A(y) = \pi((4y)^2 - (y^3)^2) = \pi(16y^2 - y^6)$$

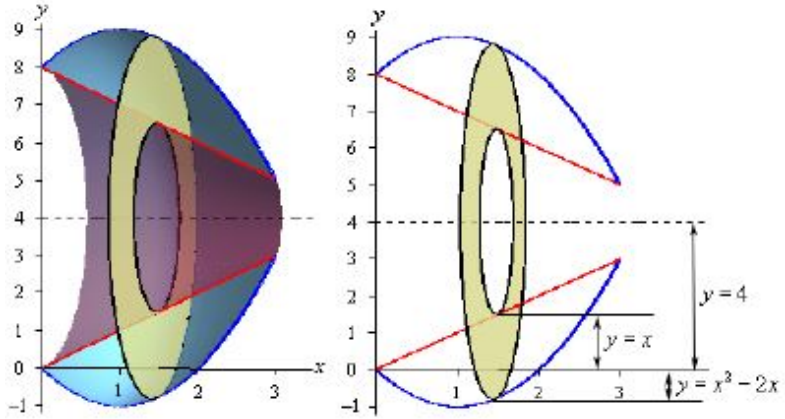
Working from the bottom of the solid to the top we can see that the first cross-section will occur at $y = 0$ and the last cross-section will occur at $y = 2$. These will be the limits of integration. The volume is then,

$$\begin{aligned} V &= \int_c^d A(y)\,dy \\ &= \pi \int_0^2 (16y^2 - y^6) \\ &= \pi \left(\frac{16}{3} y^3 - \frac{1}{7} y^7 \right)_0^2 \\ &= \frac{512\pi}{21}. \end{aligned}$$

(b) Sketches.







Inner radius $r(x) = 4 - x$ and the outer radius
$R(x) = 4 - (x^2 - 2x) = -x^2 + 2x + 4$
The cross sectional area is

$$A(x) = \pi((-x^2 + 2x + 4)^2 - (4 - x)^2) = \pi(x^4 - 4x^3 - 5x^2 + 24x)$$

We also have $0 \leqslant x \leqslant 3$. The volume is then,

$$\begin{aligned} V &= \int_a^b A(x)\,dx \\ &= \pi \int_0^3 (x^4 - 4x^3 - 5x^2 + 24x)\,dx \\ &= \pi \left(\frac{1}{5}x^5 - x^4 - \frac{5}{3}x^3 + 12x^2 \right)_0^3 = \frac{153\pi}{5}. \end{aligned}$$

(c) - (e) class exercise.

The Shell/Cylinder Method

Theorem 4

(1) **Shell Formula for Revolution About a Vertical Line**

The volume of the solid generated by revolving the region between the x-axis and the graph of a continuous function $y = f(x) \geqslant 0$, $L \leqslant a \leqslant x \leqslant b$, *about a vertical line* $x = L$ *is*

$$V = 2\pi \int_a^b (\textit{Shell radius})(\textit{Shell height})\, dx \tag{5}$$

(2) **Shell Formula for Revolution About a Horizontal Line**

$$V = 2\pi \int_c^d (\textit{Shell radius})(\textit{Shell height})\, dy \tag{6}$$

Example 3

(a) Determine the volume of the solid obtained by rotating the region bounded by $y = (x-1)(x-3)^2$ and the $x-$axis about the $y-$axis.

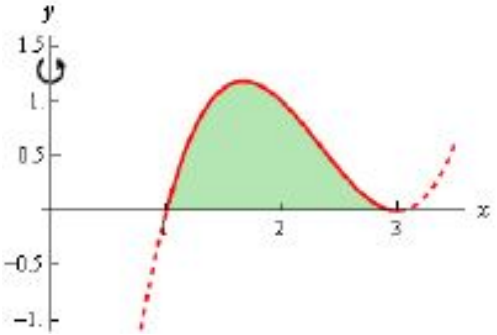
(b) Determine the volume of the solid obtained by rotating the region bounded by $y = \sqrt[3]{x}$, $x = 8$ and the $x-$axis about the $x-$axis.

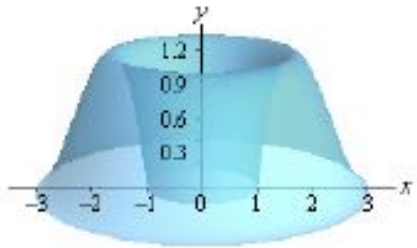
(c) Determine the volume of the solid obtained by rotating the region bounded by $y = 2\sqrt{x-1}$ and $y = x - 1$ about the line $x = 6$.

(d) Determine the volume of the solid obtained by rotating the region bounded by $x = (y-2)^2$ and $y = x$ about the line $y = -1$.

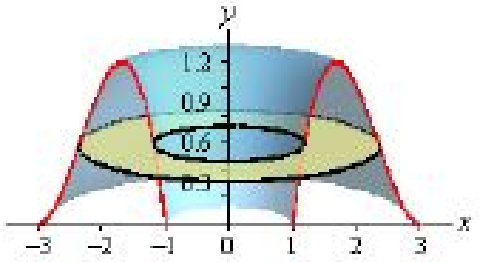
Solution

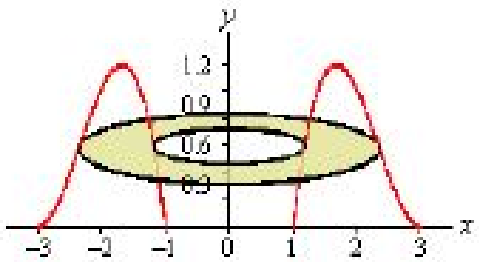
(a) Sketch





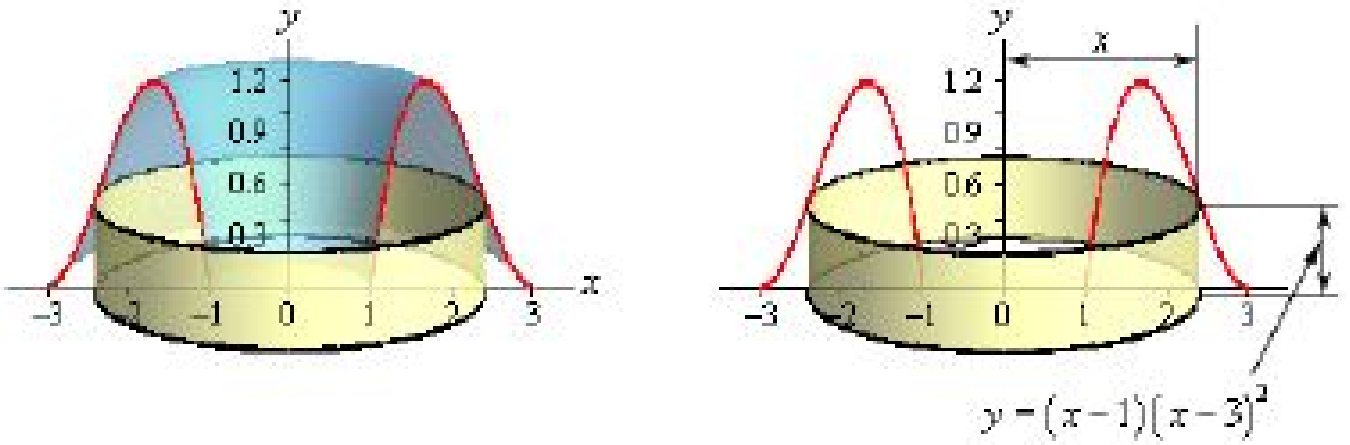
If we were to use washers/rings on this solid here is what a typical ring would look like.





This leads to several problems. First, both the inner and outer radius are defined by the same function. This usually means more work than other methods, so its often not the best approach.

What we need to do is to find a different way to cut the solid that will give us a cross-sectional area that we can work with. One way to do this is to think of our solid as a lump of cookie dough and instead of cutting it perpendicular to the axis of rotation we could instead senter a cylindrical cookie cutter on the axis of rotation and push this down into the solid. Doing this would give the following picture.



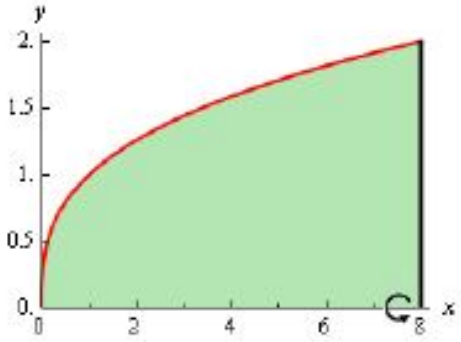
Doing this gives us a cylinder or shell in the object and we can easily find its surface area. The surface area of this cylinder is,

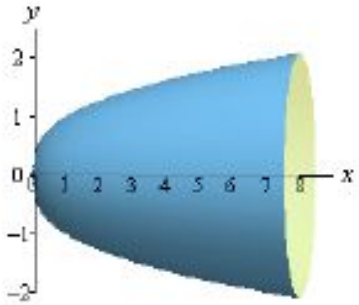
$$\begin{aligned} A(x) &= 2\pi(\text{Shell radius})(\text{Shell height}) \\ &= 2\pi(x)((x-1)(x-3)^2) \\ &= 2\pi(x^4 - 7x^3 + 15x^2 - 9x) \end{aligned}$$

The first cylinder will cut into the solid at $x = 1$ and as we increase x to $x = 3$ we will completely cover both sides of the solid since expanding the cylinder in one direction will automatically expand it in the other direction as well. The volume of this solid is then,

$$\begin{aligned} V &= \int_a^b A(x)\,dx \\ &= 2\pi \int_1^3 (x^4 - 7x^3 + 15x^2 - 9x)\,dx \\ &= 2\pi \left(\frac{1}{5}x^5 - \frac{7}{4}x^4 + 5x^3 - \frac{9}{2}x^2 \right)_1^3 \\ &= \frac{24\pi}{5}. \end{aligned}$$

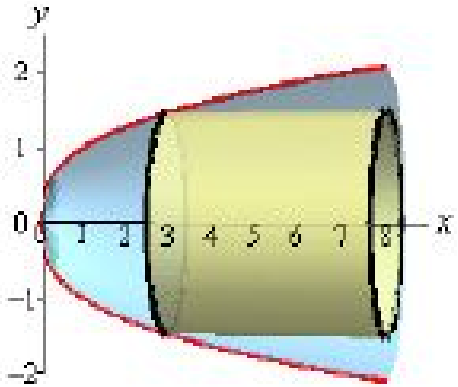
(b) Sketch

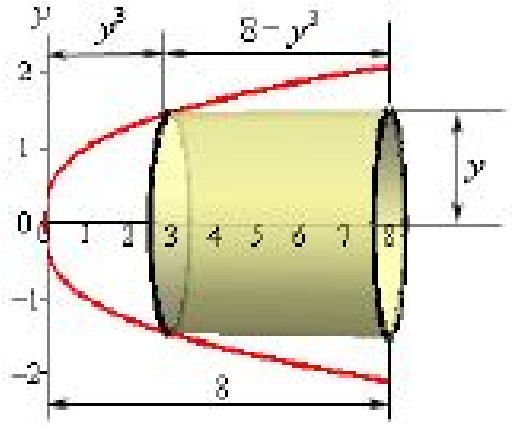




We are rotating about a horizontal axis. This means that the area will be a function of y and so our equation will also need to be written in $x = f(y)$ form.

$$y = \sqrt[3]{x} \quad \Rightarrow \quad x = y^3.$$





The distance from the edge out to the line is $x = 8$ and so the width is then $8 - y^3$.

The cross-sectional area in this case is,

$$\begin{aligned} A(y) &= 2\pi(\text{Shell radius})(\text{Shell height}) \\ &= 2\pi(y)(8-y^3) \\ &= 2\pi(8y-y^4). \end{aligned}$$

The first cylinder will cut into the solid at $y = 0$ and the final cylinder will cut in at $y = 2$ and so these are our limits of integration.
The volume of this solid is,

$$\begin{aligned} V &= \int_c^d A(y)\,dy \\ &= 2\pi \int_0^2 (8y-y^4)\,dx \\ &= 2\pi \left(4y^2 - \frac{1}{5}y^5\right)_0^2 \\ &= \frac{96\pi}{5}. \end{aligned}$$

(c) - (d) class exercise.